

E-Commerce User Interaction Dashboard

Business Intelligence Report — DSA3050A Mid-Semester Exam · Kaggle Dataset · 100K rows × 7 columns · Jan 2023 – May 2026

100K

Total Interactions

6,944

Unique Users

967

Unique Products

23.5%

View-to-Cart Rate

17.9s

Avg Dwell Time

10.2%

Wishlist Rate



Key Business Insights

Insight 1: Strong View-to-Cart Conversion — 23.5%

Far above the 3–5% benchmark. Product ID **6e9b539a** drove 6,777 interactions (~7% of all activity) — prime for promotional spotlight.

Insight 2: Accelerating Platform Growth — 3× Traffic Increase

Monthly interactions grew from ~450 (Jan 2023) to 3,500+ by Apr–May 2026 — a **6× increase**. YoY growth 2023→2024 was **52%**. Q1/Q2 consistently strongest.

Insight 3: 45% Are 'Casual Visitors' — Major Retention Opportunity

45,001 interactions (45%) from Casual Visitors vs. only 7,854 (7.9%) High Intent Buyers. **7,526 interactions (7.5%)** show Churn Risk — abandoned carts/wishlists warrant re-engagement flows.

Data Transformations & Dashboard

Power Query Transformations

- Renamed columns, typed data, removed duplicates/blanks, cleaned Interaction Type
- Split IDs, merged User-Session Key, computed Dwell Time (ms→s)
- Engagement Level, date parts, filtered View/Click >5s, sorted, indexed
- Group By, Pivot, Date Table, nested Customer Category (5 segments)

Dashboard Visuals & Interactivity

- KPI cards, bar/column/line charts, donut, matrix, table, top 10 products, engagement distribution
- Slicers: Year, Interaction Type, Engagement Level
- Drill-down: Year → Quarter → Month; cross-filtering enabled